An Ordered Poisson Coupling for Mittag–Leffler Laws

Exact stochastic comparison, optimal scale, and a quantitative gap

Research note prepared for Keio

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Abstract

Let Z_α have Laplace transform $\mathbb{E}e^{-tZ_\alpha} = e^{-t^\alpha}$, and set $M_\alpha = Z_\alpha^{-\alpha}$, $0 < \alpha < 1$. We give a complete probabilistic proof that $M_\alpha/\Gamma(1-\alpha)$ decreases in the usual stochastic order, with optimal scale. This is the comparison proposed in Simon's 2014 Remark 9 and restated in Ferreira–Simon's 2025 Remark 7(b), with the original normalization and full index range. One Poisson arrival sequence realizes every member of the family as

$$Y_\alpha = \left(\sum_{j \geq 1} T_j^{-1/\alpha} \right)^{-\alpha}.$$

The resulting trajectories are smooth, strictly decreasing, and strictly log-concave. We also prove an explicit positive lower bound for the CDF difference at every positive threshold, and a stronger bound linear in the index difference on compact index intervals. The Poisson-series and norm principles are classical and are credited as such. Historical priority of the specific results remains unverified. This proof draft is not presented as a certified success under all six novelty requirements recorded in Appendix A.

1 Original statement and sources

The stochastic-order clause was already conjectured by T. Simon, *Comparing Fréchet and positive stable laws*, Electronic Journal of Probability 19 (2014), no. 16, 1–25, Theorem 3, equation (5.4), and Remark 9; [1]. The arXiv v2 is dated 27 January 2014. It conjectures that for all $0 < \beta < \alpha < 1$,

$$\Gamma(1-\beta)M_\alpha \prec_{\mathrm{st}} \Gamma(1-\alpha)M_\beta,$$

where $X \prec_{\mathrm{st}} Y$ means $X \leq_{\mathrm{st}} Y$ and no $c > 1$ satisfies $cX \leq_{\mathrm{st}} Y$. The source proves a restricted parameter region only. The unrestricted convex-order clause next to it is a different assertion and is not counted as a new result here. R. A. C. Ferreira and T. Simon, *Mittag–Leffler functions and convex ordering*, arXiv:2512.04940v1, 4 December 2025, Remark 7(b), propose that

$$\alpha \longmapsto \frac{M_\alpha}{\Gamma(1-\alpha)}$$

is non-increasing for the usual stochastic order on $(0,1)$. Here $M_\alpha = Z_\alpha^{-\alpha}$ in distribution, where $\mathbb{E}e^{-tZ_\alpha} = e^{-t^\alpha}$ for $t \geq 0$. The current arXiv record, checked on 8 November 2026, lists only v1 [2]. The author's current publications page also lists that preprint. The original claim, normalization, index interval, and probability order are retained exactly below.

2 Main result and classical inputs

The Poisson Laplace functional and monotonicity of ℓ^p norms are classical. The series used below is the positive LePage series. None of these ingredients is claimed to be new. The proposed contribution is their simultaneous, correctly normalized application to the specific stochastic-order conjecture, together with the explicit separation bound. Poisson representations through ℓ^p norms also occur explicitly in Oesting's max-stable constructions [5]; that viewpoint is credited as prior work. Janßen's Corollary 6.3 [6], reproducing a result of Schlather, also identifies stable-power laws as limits of sample norms. These are close precedents for the qualitative mechanism. Standard Poisson references are [3, 4]. Priority of the specific comparison or the quantitative estimates has not been established by the bounded search described in Appendix A.

Theorem 2.1 (Full comparison and exact optimal scale)**.** *There is a coupling $(Y_\alpha)_{0<\alpha<1}$ with*

$$Y_\alpha \overset{d}{=} \frac{M_\alpha}{\Gamma(1-\alpha)}$$

and one event of probability one on which

$$Y_\beta < Y_\alpha \qquad \text{for every } 0 < \alpha < \beta < 1.$$

For every such pair and every $x > 0$,

$$\mathbb{P}(Y_\beta \le x) > \mathbb{P}(Y_\alpha \le x).$$

Furthermore, for $c > 0$,

$$cY_\beta \le_{\mathrm{st}} Y_\alpha \quad \Longleftrightarrow \quad c \le 1.$$

Equivalently, the largest c for which $cM_\beta \le_{\mathrm{st}} M_\alpha$ is $\Gamma(1-\alpha)/\Gamma(1-\beta)$.

The following sections prove every clause. In particular, strictness at every threshold is proved by an explicit positive-probability event, not inferred from strictness of a coupling alone.

3 Construction and marginal identification

Let $0 < T_1 < T_2 < \cdots$ be the arrivals of one rate-one Poisson process, and use this same sequence for every $0 < \alpha < 1$. On the probability-one event $T_j/j \to 1$, define

$$S_\alpha = \sum_{j \ge 1} T_j^{-1/\alpha}, \qquad Y_\alpha = S_\alpha^{-\alpha}.$$

Every series converges on this one event, simultaneously for all $\alpha \in (0,1)$: for each such α, its tail is bounded by a constant times $\sum j^{-1/\alpha}$. For $t \ge 0$, the Poisson Laplace functional gives

$$\begin{aligned} \mathbb{E}e^{-tS_\alpha} &= \exp\left(-\int_0^\infty (1 - e^{-tu^{-1/\alpha}})\, du\right) \\ &= \exp\left(-\alpha t^\alpha \int_0^\infty (1 - e^{-v}) v^{-\alpha-1}\, dv\right) \\ &= \exp\bigl(-\Gamma(1-\alpha) t^\alpha\bigr). \end{aligned}$$

The substitution is $v = tu^{-1/\alpha}$ and the last equality follows by integration by parts; both boundary terms vanish for $0 < \alpha < 1$. Thus

$$S_\alpha \overset{d}{=} \Gamma(1-\alpha)^{1/\alpha} Z_\alpha, \qquad Y_\alpha \overset{d}{=} \frac{M_\alpha}{\Gamma(1-\alpha)}.$$

The Poisson Laplace functional here can be derived first for simple nonnegative functions supported in a finite interval from independence of the Poisson counts, then passed to this integrand by monotone convergence. Consequently no distributional assertion about the LePage series has to be imported separately.

4 Pathwise order

Write $a_j = T_j^{-1}$. Then

$$Y_\alpha = \|a\|_{1/\alpha}^{-1}.$$

If $0 < \alpha < \beta < 1$, set $p = 1/\alpha > q = 1/\beta > 1$. Normalize $b_j = a_j/\|a\|_q$. Every b_j lies in $(0,1)$ and $\sum b_j^q = 1$, so $\sum b_j^p < 1$. Therefore

$$Y_\beta < Y_\alpha \quad \text{almost surely.}$$

This proves the original usual stochastic order, with all index pairs covered by a single coupling and one probability-one event. Here $X \le_{\mathrm{st}} W$ means $\mathbb{P}(X > x) \le \mathbb{P}(W > x)$ for every real x. The identity with a norm supplies the representation; the strict norm inequality supplies the actual comparison.

5 Optimality in Simon's original sense

For $0 < s < 1$, the fractional-moment integral for a nonnegative random variable yields

$$\mathbb{E}Z_\alpha^{\alpha s} = \frac{\alpha s}{\Gamma(1-\alpha s)} \int_0^\infty (1 - e^{-t^\alpha}) t^{-\alpha s - 1}\, dt = \frac{\Gamma(1-s)}{\Gamma(1-\alpha s)}.$$

The substitution $u = t^\alpha$ and integration by parts justify the last step. Thus

$$\mathbb{E}Y_\alpha^{-s} = \frac{\Gamma(1-\alpha)^s \Gamma(1-s)}{\Gamma(1-\alpha s)}, \qquad \lim_{s \uparrow 1} \frac{\mathbb{E}Y_\beta^{-s}}{\mathbb{E}Y_\alpha^{-s}} = 1.$$

If $cY_\beta \le_{\mathrm{st}} Y_\alpha$ held with $c > 1$, applying the decreasing function $y \mapsto y^{-s}$ would give $c^{-s}\mathbb{E}Y_\beta^{-s} \ge \mathbb{E}Y_\alpha^{-s}$. Letting $s \uparrow 1$ gives $c^{-1} \ge 1$, a contradiction. Consequently $Y_\beta \prec_{\mathrm{st}} Y_\alpha$, which is exactly Simon's original optimal order after positive rescaling.

6 An explicit positive gap at every threshold

Fix $0 < \alpha < \beta < 1$ and $x > 0$. Set

$$u = x2^\alpha, \qquad v = x2^\beta, \qquad a = \frac{2u+v}{3}, \qquad b = \frac{u+2v}{3}, \qquad A = x^{-1/\alpha} - 2a^{-1/\alpha} > 0.$$

Choose, for example,

$$R = 1 + \max\left\{b, \left(\frac{2\alpha}{(1-\alpha)A}\right)^{\alpha/(1-\alpha)}\right\}.$$

The expected tail of S_α beyond R is

$$\mathbb{E}\sum_{T_j > R} T_j^{-1/\alpha} = \frac{\alpha}{1-\alpha} R^{1-1/\alpha} < A/2.$$

By Markov's inequality the tail is less than A with probability greater than $1/2$. Independently, the event that there are exactly two points in $[a, b]$ and no other points in $[0, R]$ has probability $e^{-R}(b-a)^2/2$. On the intersection,

$$S_\alpha < 2a^{-1/\alpha} + A = x^{-1/\alpha}, \qquad S_\beta \geq 2b^{-1/\beta} > x^{-1/\beta}.$$

Hence $Y_\beta < x < Y_\alpha$ there, and the common coupling yields the explicit bound

$$\mathbb{P}(Y_\alpha > x) - \mathbb{P}(Y_\beta > x) \geq \frac{e^{-R}(b-a)^2}{4} > 0.$$

This closes the stronger statement of strict stochastic separation for every positive threshold, without relying on the generally invalid inference that a strict coupling alone implies strict tail inequalities at every threshold.

7 Regularity, entropy, and curvature

For a fixed sample on $T_j/j \to 1$, the series and all its parameter derivatives converge locally uniformly on $(0, 1)$. Indeed, on a compact index interval with upper endpoint $c < 1$, they are dominated by a constant times $j^{-1/c}(1 + \log j)^k$ for each fixed derivative order. Let

$$p_j(\alpha) = \frac{T_j^{-1/\alpha}}{S_\alpha}, \qquad H(\alpha) = -\sum_j p_j(\alpha) \log p_j(\alpha).$$

Direct differentiation gives

$$\frac{d}{d\alpha} \log Y_\alpha = -H(\alpha) < 0, \qquad \frac{d^2}{d\alpha^2} \log Y_\alpha = -\frac{\mathrm{Var}_{p(\alpha)}(\log T_j)}{\alpha^3} < 0.$$

All sums are finite by the same domination. Strictness holds since infinitely many weights are positive and the arrival times are distinct. The coupling therefore has smooth, strictly decreasing, strictly log-concave trajectories on $(0, 1)$. This is a statement about sample paths in the chosen coupling, not about log-concavity of the marginal density in its spatial variable.

For an explicit finite-parameter identity, set $r = \alpha/\beta \in (0, 1)$. Then

$$\frac{Y_\beta}{Y_\alpha} = \left(\sum_j p_j(\alpha)^r \right)^{-\beta} = \exp\{-(\beta - \alpha) H_r(p(\alpha))\}, \qquad H_r(p) = \frac{\log \sum_j p_j^r}{1 - r}.$$

Indeed, $S_\beta = S_\alpha^r \sum_j p_j(\alpha)^r$. The sum is finite because both S_α and S_β are finite. Strict concavity also yields, for $0 < \theta < 1$,

$$Y_{\theta\alpha + (1-\theta)\beta} > Y_\alpha^\theta Y_\beta^{1-\theta}.$$

8 Endpoints and Mittag–Leffler consequence

The standard ℓ^p limit gives $Y_\alpha \to T_1$ as $\alpha \downarrow 0$. As $\alpha \uparrow 1$, for every fixed k the norm is bounded below by $(\sum_{j=1}^k T_j^{-1/\alpha})^\alpha \to \sum_{j=1}^k T_j^{-1}$. Since $T_j/j \to 1$, the latter partial sums diverge, proving $Y_\alpha \to 0$ almost surely. Consequently, for $x > 0$ and $0 < \alpha < \beta < 1$,

$$\frac{1}{1+x} < E_\alpha\left(-\frac{x}{\Gamma(1-\alpha)}\right) < E_\beta\left(-\frac{x}{\Gamma(1-\beta)}\right) < 1.$$

Here $E_\alpha(z) = \sum_{n\ge 0} z^n/\Gamma(1+\alpha n)$. The equality of its Laplace representation to that of M_α follows from the negative moments of Z_α:

$$\mathbb{E}Z_\alpha^{-r} = \frac{1}{\Gamma(r)}\int_0^\infty t^{r-1}e^{-t^\alpha}\,dt = \frac{\Gamma(1+r/\alpha)}{\Gamma(1+r)} \quad (r>0).$$

At $r=\alpha n$, these moments give $\mathbb{E}M_\alpha^n = n!/\Gamma(1+\alpha n)$. For $t\ge 0$, Tonelli identifies $\mathbb{E}e^{tM_\alpha}$ with the defining series $E_\alpha(t)$, which is finite by Stirling's formula. Domination by $e^{|z|M_\alpha}$ then justifies the power-series calculation for all complex z, including the negative arguments used above. The strict lower comparison uses $Y_\alpha < T_1$, since $S_\alpha > T_1^{-1/\alpha}$. The endpoint limits in the Laplace inequality follow by bounded convergence. The value at $x=0$ is equality throughout.

9 A locally uniform linear separation estimate

The preceding two-point event proves a positive gap for every pair. The next result is stronger when the indices are close: its lower bound is linear in their separation on each fixed compact index interval.

Theorem 9.1 (Linear CDF separation)**.** *Fix $0<a<b<1$ and $x>0$. For every $a\le\alpha<\beta\le b$, write $F_\gamma(x)=\mathbb{P}(Y_\gamma\le x)$. Define*

$$C = 1+\max\left\{2^b, \left(\frac{8xb}{1-b}\right)^{b/(1-b)}\right\}, \qquad \eta = \left(\frac{x}{Cx+1}\right)^{1/a}.$$

Then $0<\eta<1/2$ and

$$F_\beta(x)-F_\alpha(x) \ge (\beta-\alpha)\frac{xe^{-Cx}}{4}\log\frac{1}{1-\eta} > 0.$$

The constant is uniform over every pair in the chosen compact index interval. In particular, unlike the preceding two-point-event estimate, it remains linear as $\beta-\alpha$ decreases to zero.

Proof. Use independent rate-one exponential variables $E_1, E_2, \ldots$, and put $V_1 = 0$, $V_j = E_2+\cdots+E_j$ for $j\ge 2$. The positive V_j form a rate-one Poisson process independent of E_1. Conditioning on all the V_j, define for $t>0$

$$S_\gamma(t) = \sum_{j\ge 1}(t+V_j)^{-1/\gamma}, \qquad Y_\gamma(t) = S_\gamma(t)^{-\gamma}.$$

Almost surely $V_j/j\to 1$. The sums and all parameter derivatives converge locally uniformly on $(0,1)\times(0,\infty)$, by comparison with $j^{-1/b'}(1+\log j)^k$ on any compact index range ending at $b'<1$. The map $t\mapsto Y_\gamma(t)$ is smooth and strictly increasing. It tends to zero as $t\downarrow 0$, since $Y_\gamma(t)\le t$. It tends to infinity as $t\to\infty$, since $S_\gamma(t)\to 0$ by dominated convergence, using the summable sequence $(1+V_j)^{-1/\gamma}$ for $t\ge 1$. Consequently a unique smooth function $\tau_\gamma(x)>0$ satisfies $Y_\gamma(\tau_\gamma(x))=x$, and conditioning gives

$$F_\gamma(x) = 1-\mathbb{E}e^{-\tau_\gamma(x)}.$$

At $t=\tau_\gamma(x)$, let

$$p_j = \frac{(t+V_j)^{-1/\gamma}}{S_\gamma(t)}, \qquad H = -\sum_j p_j\log p_j, \qquad D = \sum_j \frac{p_j}{t+V_j}.$$

The previously proved entropy derivative and direct differentiation in t give $\partial_\gamma \log Y = -H$ and $\partial_t \log Y = D$. The implicit-function theorem therefore gives

$$\frac{d}{d\gamma}\tau_\gamma(x) = \frac{H}{D} \geq \tau_\gamma(x)H,$$

because $D \leq 1/t$.

Consider the event

$$G = \left\{ V_2 \leq 1, \quad \sum_{j\geq 2}(C + V_j/x)^{-1/b} \leq 1 - C^{-1/b} \right\}.$$

Campbell's formula gives

$$\mathbb{E}\sum_{j\geq 2}(C + V_j/x)^{-1/b} = \frac{xb}{1-b}C^{1-1/b} < \frac{1}{8}.$$

The threshold $1 - C^{-1/b}$ exceeds $1/2$. Markov's inequality therefore bounds the failure probability of the second condition by $1/4$. Since $\mathbb{P}(V_2 \leq 1) = 1 - e^{-1} > 1/2$, the union bound gives $\mathbb{P}(G) > 1/4$; no independence between the two conditions defining G is assumed.

On G, uniformly for $\gamma \in [a, b]$, the fact that $C + V_j/x > 1$ implies

$$x^{1/\gamma}S_\gamma(Cx) = C^{-1/\gamma} + \sum_{j\geq 2}(C + V_j/x)^{-1/\gamma} \leq C^{-1/b} + \sum_{j\geq 2}(C + V_j/x)^{-1/b} \leq 1.$$

Thus $Y_\gamma(Cx) \geq x$. Together with $Y_\gamma(t) \leq t$, this shows $x \leq \tau_\gamma(x) \leq Cx$. At the threshold root,

$$p_1 = \left(\frac{x}{\tau_\gamma(x)}\right)^{1/\gamma} \geq \eta, \qquad p_2 = \left(\frac{x}{\tau_\gamma(x) + V_2}\right)^{1/\gamma} \geq \eta.$$

Every p_j is therefore at most $1 - \eta$, and hence $H \geq \log(1/(1-\eta))$. It follows on G that

$$e^{-\tau_\gamma(x)}\frac{d}{d\gamma}\tau_\gamma(x) \geq xe^{-Cx}\log\frac{1}{1-\eta},$$

Integrate pathwise from α to β, restrict expectation to G, and use nonnegativity on the complement. This proves the claimed bound. No differentiation under an expectation is required. □

10 A stronger endpoint normalization and a closed-form check

Proposition 10.1. *In the same coupling, almost surely*

$$\lim_{\alpha\uparrow 1}\frac{Y_\alpha}{1-\alpha} = 1.$$

Consequently $\Gamma(1-\alpha)Y_\alpha \to 1$ *almost surely.*

Proof. Put $p = 1/\alpha > 1$. For any $\varepsilon \in (0, 1)$, the event $T_j/j \to 1$ supplies J such that $(1-\varepsilon)j \leq T_j \leq (1+\varepsilon)j$ for $j \geq J$. Integral comparison gives $(p-1)\sum_{j\geq J} j^{-p} \to 1$ as $p \downarrow 1$. The finite prefix vanishes after multiplication by $p - 1$. Squeezing and then letting $\varepsilon \downarrow 0$ proves

$$(p-1)\sum_j T_j^{-p} \longrightarrow 1.$$

It follows that

$$\log\frac{Y_\alpha}{1-\alpha} = (\alpha-1)\log(1-\alpha) - \alpha\log\alpha + o(1) \longrightarrow 0.$$

Finally $(1-\alpha)\Gamma(1-\alpha) \to 1$. □

At $\alpha = 1/2$, the density of $Y_{1/2}$ is

$$f_{1/2}(y) = e^{-\pi y^2/4}, \qquad y > 0.$$

Its integral is one and its mean is $2/\pi$, consistent with the moment formula. Completing the square in its Laplace integral yields

$$\mathbb{E}e^{-xY_{1/2}} = e^{x^2/\pi}\operatorname{erfc}(x/\sqrt{\pi}) = E_{1/2}(-x/\sqrt{\pi}).$$

The last identity follows also by separating the even and odd terms of the Mittag–Leffler series. This check is not an input to any all-parameter comparison.

11 Research transition and limits of transfer

The supplied corpus [7] motivated a search through positivity, complete monotonicity, and stable-law factorizations. Its items [116], [122], and [127] concern Le Roy positivity, stable-law comparisons, and Kanter factors. None of their claimed theorems is a premise of the proof here.

The useful auxiliary construction problem was to realize all the original laws on one space, preserve their exact normalization, and force a positive chance of crossing every fixed threshold. The common Poisson sequence solves the first task. Norm comparison and the conditional first-arrival estimate solve the latter tasks. This construction is applied to the historical comparison itself; an auxiliary definition is not substituted for the original target.

The positive bound of Theorem 9.1 is a substantive estimate for the original CDFs. Its proof combines a conditional inverse threshold, an entropy derivative, Campbell's formula, and Markov's inequality. These ingredients are standard. Their historical novelty in this combination remains a separate question.

No estimate for the Riemann hypothesis, the prime-circle global energy bound, or the Lorch–Muldoon all-orders resultant problem is proved here. For example, a reciprocal-prime sequence can have decreasing ℓ^p norms, but it is not a Poisson arrival sequence. Assigning the stable marginal law to it would require an additional theorem. Norm monotonicity does not supply the missing arithmetic cancellation bound.

A Evidence and the six success conditions

Mathematical validity and historical priority are separate. The statements above have complete proofs and were checked independently within this research session. They have not undergone external peer review or formal proof-assistant verification.

Gate	Requirement	Assessment
G1	Pre-existing identified problem	Verified from Simon's 2014 Remark 9 and Ferreira–Simon's 2025 Remark 7(b). These author-and-location identifiers are pre-existing; no universally standardized proper name is assumed.
G2	Original statement unchanged	The exact variables, gamma normalization, probability order, and full open index interval are retained. Equality in law and the optimal scale are proved.
G3	Actually unresolved	The cited 2014 and 2025 sources explicitly leave the general comparison open. This establishes the historical open question there, not the absence of every intervening resolution worldwide.

Gate	Requirement	Assessment
G4	Unconditional complete proof	All parameters, thresholds, and scale factors are treated. No numerical verification or unproved hypothesis is used. The mathematical argument passed independent internal checks.
G5	More than a reformulation	Norm inequalities, scale optimality, and explicit CDF separation are proved, including a locally uniform linear bound. The base principles are classical; a historically new general principle has not been established.
G6	No essentially identical prior solution	Unverified. Bounded searches and inspection of close precedents did not locate an identical resolution. Search absence is not a proof of priority.

Overall assessment. This is a complete proof draft for the stated stochastic comparison, with quantitative extensions. It is not a certified all-G1–G6 historic-breakthrough claim.

A.1 Primary sources and limits of the review

1. Simon [1]: journal publication and arXiv v2, especially the definition of optimal stochastic order, Theorem 3, and Remark 9. Its other stable-power family and its convex-order comparison are not confused with the target.

2. Ferreira–Simon [2]: full text, Remark 7(b), and current arXiv submission history. The latest record inspected still lists the December 2025 v1.

3. LePage–Woodroofe–Zinn [3]: bibliographic identification of the classical stable-series lineage. The exact marginal law used here is derived directly, so no uninspected variant of their theorem is required.

4. Last–Penrose [4]: the Laplace functional and Campbell formula used for the probability estimates.

5. Oesting [5], Sections 2–3: the prior Poisson ℓ^p viewpoint and the stable marginal in max-stable representations. This is a close methodological precedent. This review does not exhaust all subsequent uses of that viewpoint.

6. Janßen [6], Section 6 and Corollary 6.3: stable-power limits of positive sample norms. Combined with norm monotonicity, such limits offer another route to investigate the qualitative comparison. The present review therefore makes no claim that the underlying norm principle is historically new.

Searches included exact article identifiers and combinations of author names with stochastic order, LePage, Poisson coupling, Mittag–Leffler, and stable norms. Author publication records were inspected. The searches establish a bounded review, not worldwide novelty. The appropriate next independent evaluation is to check the exact comparison and quantitative estimates against the stable-law and Poisson-series literature.

A.2 Correction to the earlier theta-kernel research record

A preceding note in this session proved infinitely many strict center failures of odd derivative Laguerre inequalities for the Riemann Xi kernel while leaving priority unverified. The present review located Koide's preprint dated 15 August 2026 [8]. Its Theorem 1.1 states a quantitative count already implying infinitely many such failures. The qualitative conclusion must therefore be removed from any list of possible first resolutions produced in this session. This records the existence and scope of a prior claim; it does not certify every proof in that preprint.

B The decisive comparison in compact form

Take one sequence of rate-one Poisson arrivals and define

$$X_\alpha = \left(\sum_j T_j^{-1/\alpha} \right)^{-\alpha}.$$

The strong law makes all sums finite simultaneously for $0 < \alpha < 1$. The Poisson Laplace functional gives

$$\mathbb{E} \exp\left(-t \sum_j T_j^{-1/\alpha} \right) = e^{-\Gamma(1-\alpha)t^\alpha}.$$

Thus X_α has the exact law $M_\alpha/\Gamma(1-\alpha)$. For every realization, X_α is the reciprocal $\ell^{1/\alpha}$ norm of the same positive sequence $(1/T_j)$. These reciprocals strictly decrease with α, proving the original usual stochastic order. The negative-moment limit proves optimal scale; Section 9 gives the quantitative bound beyond the basic comparison.

References

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[3] R. LePage, M. Woodroofe, and J. Zinn, *Convergence to a stable distribution via order statistics*, Annals of Probability **9** (1981), no. 4, 624–632. doi:10.1214/aop/1176994367.

[4] G. Last and M. Penrose, *Lectures on the Poisson Process*, Cambridge University Press, 2017. Author manuscript: https://stoch.math.kit.edu/img/Last/lastpenrose2017.pdf.

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[6] A. Janßen, *Limit laws for power sums and norms of i.i.d. samples*, Probability Theory and Related Fields **146** (2010), 515–533. doi:10.1007/s00440-008-0198-y.

[7] QEEDDD, *Verifiable Mathematical Reasoning Corpus*, snapshot inspected 8 November 2026, commit c30d40119134a59eda6fa5cc6331f27c8e67d034. https://github.com/QEEDDD/Verifiable-Mathematical-Reasoning-Corpus. Research drafts; their claims are not premises of this proof.

[8] A. Koide, *Power-Saving Center-Failure Counts for the Derivative Laguerre Inequalities of the Riemann Xi Kernel*, preprint, 15 August 2026. doi:10.5281/zenodo.21951273.